ABSTRACTS

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Can a surgeon wear a monitor: physical attributes of an extended reality (XR) near-eye headset

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Introduction: When performing image guided procedures, surgeons need to look at one or more monitors. A heads-up display (HUD) designed with XR technology could offer the surgeons the opportunity to look directly at the operative field and their hands during the procedure. While the potential to improve hand-eye coordination is attractive, the XR HUD would also need to either replicate or exceed the visual acuity of current operating room monitors. We designed a system to meet these expectations.

Methods: The following attributes were considered necessary for the design of the HUD. 1) high transparency stereoscopic reflective waveguides to receive the video images; 2) large field of view to give surgeon the same visual experience when looking at a monitor; 3) adequate resolution and brightness, 4) high definition miniature projectors to display the images onto the lenses, and 5) software designed to arrange image position on the lenses, to generate real-time 2D/3D images and to control image opacity so the surgeons could look simultaneously at the operative site and the projected image (i.e. live ultrasound, fluoroscopic or endoscopic). The Normalized Angular Resolution (NAR) metric was used to evaluate display performance by considering both pixel distinguishability and resolution in relation to human vision for the HUD vs. a standard OR monitor. For 4K monitors positioned at least 7" from the surgeons, the NAR demonstrated consistent power-law behavior, expressed as: $-1.759\ y = 349,174{,}716 * x$ where y is the total effective pixels. This relationship provided crucial insights into long-term performance characteristics and helped predict system behavior across various surgical scenarios.

Results: A near-eye stereoscopic wearable display was developed with see-through optics which were housed in a 3D-printed frame (figure 1). The specifications of the lenses included: a fixed viewing distance (20 mm), resolution of 1920 x 1080 (2,073,600 pixels), dimensions 13 x 10 mm, pixel density 10,290,798.4 pixels/in², bit-depth 24 for RGB and for grayscale 8. The angular resolution of 1.02 arc minute/pixel (0.017°/pixel) closely matched human visual acuity. The graph shows superior color and greyscale depth advantage across almost the entire distance domain (figures 2,3). Beyond 11.5" the near-eye display exceeds resolution and color acuity of current OR monitors.

Conclusion: Our near-eye display system demonstrated measurable advantages in medical visualization through both theoretical analysis and practical validation. Our research established clear performance thresholds through intersection points at parity, and order of magnitude improvement for RGB and Greyscale images with the key thresholds being clinically significant is 11.5", 40"and 75" respectively.

Figure 1: Resident wears HUD during image-guided procedure. Figure 2&3: Solid black regression line for 4K monitor demonstrating decreased acuity at greater distances vs. no change for near-eye HUD (solid red line), RGB and greyscale respectively.

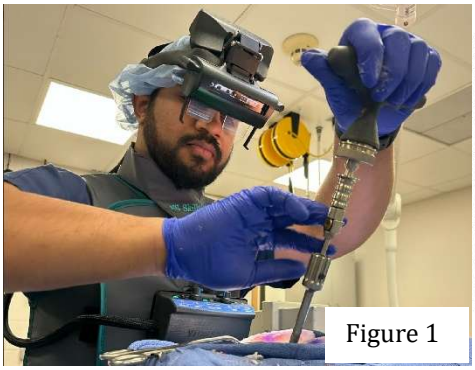
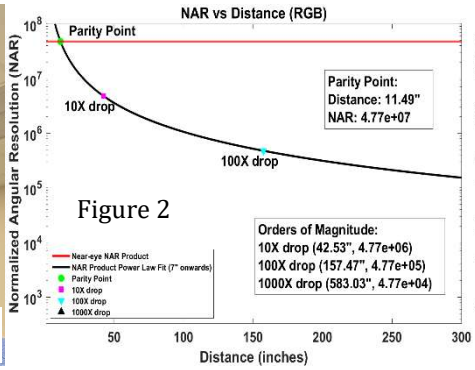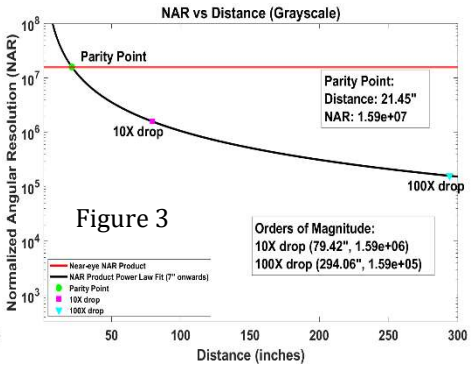
Figure 1